



Figure 1: How shc works

```
#yum install wget
#cd /usr/local/src
# wget -c http://www.datsi.fi.upm.es/~frosal/sources/shc-3.8.9b.tgz
```

Extract the tar package:

```
#tar -zxvf shc-3.8.9b.tgz
#cd shc-3.8.9b
```

To compile SHC, execute the *make* command, as follows:

```
[root@test shc-3.8.9b]# make
cc -Wall shc.c -o shc
***      Do you want to probe shc with a test script?
***      Please try... make test
```

To install the SHC binary, execute the commands below:

```
[root@test shc-3.8.9b]# make install
***      Installing shc and shc.1 on /usr/local
***      Do you want to continue? y
```

<<-----Press Y

```
install -c -s shc /usr/local/bin/
install -c -m 644 shc.1 /usr/local/man/man1/
install: target '/usr/local/man/man1/' is not a directory:
No such file or directory
make: *** [install] Error 1
```

It will show an error message, as the minimal installation default man package is not installed. You can ignore it, install the man package or create the */usr/local/man/man1/* directory, and again execute the *make install* command.

```
[root@test shc-3.8.9b]# make install
***      Installing shc and shc.1 on /usr/local
***      Do you want to continue? y
<<-----Press Y
install -c -s shc /usr/local/bin/
install -c -m 644 shc.1 /usr/local/man/man1/
```

cross check that shc is installed properly

```
[root@test shc-3.8.9b]# which shc
/usr/local/bin/shc
```

It will display the path of the SHC binary.

Hands-on usage of SHC

1. Creating a test shell script

First, create the test shell script that you are going to use to encrypt. Let's create a script for testing purposes, which will display a simple 'Welcome to the Linux world' message, as follows:

```
$vi welcome.sh
#!/bin/sh
echo "Welcome to linux world"
```

save using: *wq*

2. Encrypting using SHC

Encrypt the *welcome.sh* shell scripting using SHC, as shown below:

```
[root@test src]# shc -v -f welcome.sh
shc shll=sh
shc [-i]=-c
shc [-x]=exec '%s' "$@"
shc [-l]=
shc opts=
shc: cc welcome.sh.x.c -o welcome.sh.x
shc: strip welcome.sh.x
shc: chmod go-r welcome.sh.x
```

This will create two extra files, as shown below:

```
[root@test src]# ll welcome.sh*
-rw-r--r-- 1 root root 40 Dec 16 04:39 welcome.sh
-rwx--x--x 1 root root 11552 Dec 16 05:23 welcome.sh.x
-rw-r--r-- 1 root root 9236 Dec 16 05:23 welcome.sh.x.c
```

- *welcome.sh* is the original unencrypted shell script
- *welcome.sh.x* is the encrypted shell script in binary format
- *welcome.sh.x.c* is the C source code of the *welcome.sh* file. This C source code is compiled to create the above encrypted *welcome.sh.x* file.

3. Executing the encrypted shell script

Now, let us execute the encrypted shell script to make sure it works as expected.

```
[root@test src]# ./welcome.sh.x
Welcome to Linux World
```

Please note that the binary itself is dependent on the shell (the first line provided in *welcome.sh*, i.e., */bin/sh*) is